How Far Can We Trust Labels Copied Between Proteins?

EAT benchmark · Project 4 · Protein Prediction II – 02.11.2026



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Most proteins have never been studied in a lab

150,000,000

protein sequences in UniProt [1]

48,401

with a lab-measured enzyme function [1]

Today: copy the label of the protein with the most similar sequence.
Unreliable once fewer than ~30 % of the amino acids match.

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025); counts: UniProtKB 2026_03, queried 28.10.2026.

The idea: let a language model decide which proteins are alike

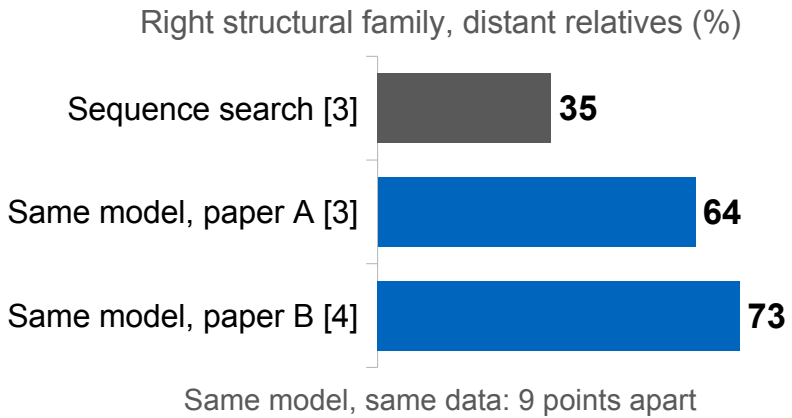


Embedding-based annotation transfer (EAT): no training, only distances

- Without any training, an early version came close to the top 10 of a worldwide prediction contest (CAFA3) [2]
- For distant relatives it finds the right structural family far more often than sequence search [3]

[2] M. Littmann … B. Rost. Embeddings transfer GO annotations beyond homology. Sci Rep (2021).
[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

Promising – but which model, and when can we trust it?



- ~12 language models, each with its own space
- Every paper sets EAT up differently
- Trust rules were set for an older model
- No tested “safe distance” exists

Our task: a fair, open benchmark

Q1 Which model and distance copy labels best – and when do they beat sequence search?

Q2 Up to which distance can we trust a copied label?

[3] Heinzinger … Rost (2022). [4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model. arXiv (2023). [5] Y. Mahlich … Y. Bromberg. HFSP. Bioinformatics (2018).

A language model turns each protein into a point in space

Protein › **Language model** › **Vector** › Nearest neighbour › Copy label

M K T A Y I A K Q R Q I S F V K …
protein = chain of amino acids

↓

Protein language model
has read millions of sequences

↓

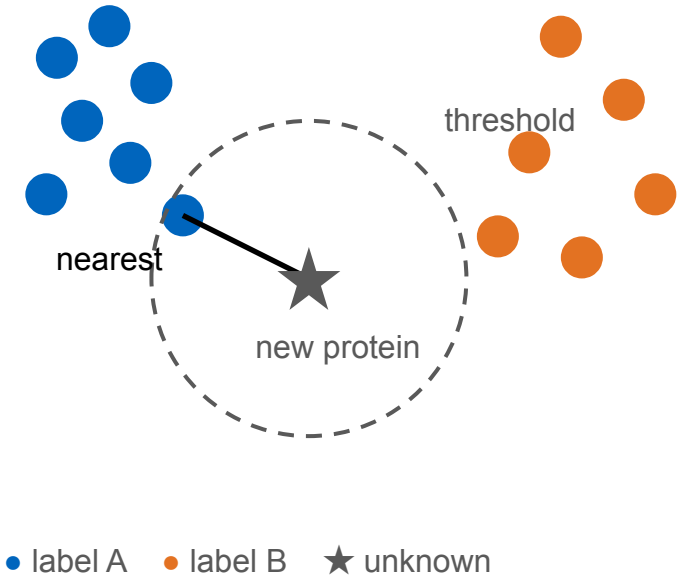
[0.12, −0.80, 0.33, …]
≈ 1,000 numbers = one point

- Learns the “grammar” of proteins, like a text model learns a language
- Similar proteins end up close together
- We compare 11 models from 4 families (ProtT5, ESM-2, ESM C, Ankh)
- **Each model builds its own space:** so each needs its own threshold
- **Nothing is trained:** vectors are computed once per model

[8] A. Elnaggar … B. Rost. ProtTrans. IEEE TPAMI (2022). [9] Z. Lin … A. Rives. ESM-2. Science (2023). [4] Elnaggar … Rost. Ankh (2023).

Copy the label of the closest protein – if it is close enough

Protein › Language model › Vector › **Nearest neighbour** › Copy label



- **Nearest neighbour:** the known protein at the smallest distance
- **Distance:** straight line (Euclidean) or angle between vectors (cosine)
- **Threshold:** too far away → do not copy the label
- **Goal:** one threshold per model, e.g. “below this distance 95 % of copied labels are correct”

[2] Littmann … Rost (2021).

Which labels? Shape (CATH) and job (EC), four levels each

Protein › Language model › Vector › Nearest neighbour › **Copy label**

CATH: the 3D shape of a domain

domain = compact, folded part of a protein
homologous = related by evolution (same ancestor)

- **3** C · Class: mixed helices and sheets
- **3.40** A · Architecture: three-layer sandwich
- **3.40.50** T · Topology: Rossmann fold
- **3.40.50.300** H · Homologous superfamily

EC: the reaction an enzyme performs

enzyme = protein that speeds up a chemical reaction

- **3** Class: hydrolase: cuts bonds with water
- **3.4** Subclass: cuts proteins
- **3.4.21** Sub-subclass: serine protease
- **3.4.21.4** Enzyme: trypsin

We measure accuracy and thresholds per level: up to which level does copying still work?

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021). [7] IUBMB. Enzyme Nomenclature (ExplorEnz).
Data: [3], [5], UniProt [1].

How we test it: reproduce, compare, scale up

Protein › Language model › Vector › Nearest neighbour › Copy label

1 Reproduce

Rerun a published result: ProtT5 on CATH

84 / 67 / 57 / 64 % per level

same numbers → our code is right

2 Compare

Copied label vs true label, per level

vs the classic way: copy from the most similar sequence

where do embeddings win?

3 Scale up

11 models × 2 distances

CATH and EC

which model, which distance, which threshold?

Fair test: tune on one part of the data, report on another

Known proteins	**Tuning set**	**Test set**
we copy labels from	choose distance, set thresholds	used once, for final numbers

[3] Heinzinger … Rost (2022): published ProtT5 result on CATH test219.

What we will deliver

1 Best model and distance (Q1)
for shape (CATH) and job (EC), and where it beats sequence similarity

2 A safe distance (Q2)
per model and level, e.g. ≥ 95 % of copied labels correct

3 One trust score (Q2)
same meaning for every model: 0.9 = 90 % of such labels correct

4 Open benchmark
code + fixed data splits: test any new model the same way

→ Thresholds go into **ProtSpace**, the lab's embedding viewer. First step (23.11): reproduce ProtT5.

Thank you – questions?

References

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025).

[2] M. Littmann … B. Rost. Embeddings from deep learning transfer GO annotations beyond homology. Sci Rep (2021).

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[4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv (2023).

[5] Y. Mahlich … Y. Bromberg. HFSP: High speed homology-driven function annotation of proteins. Bioinformatics (2018).

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021).

[7] IUBMB. Enzyme Nomenclature. ExplorEnz database, www.enzyme-database.org.

[8] A. Elnaggar … B. Rost. ProtTrans: Toward understanding the language of life. IEEE TPAMI (2022).

[9] Z. Lin … A. Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science (2023).

Project description: Chair of Bioinformatics (I12), TUM. EAT: A benchmark for annotation transfer with raw pLM embeddings (2026).